WORKED SOLUTION

STEP I 2003, Question 7

Pure • Digits and Divisibility

> **STEP I 2003, Question 7**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/03-s1-q7

The key idea

Writing $N = 100a + 10b + c$ and setting $S = N$ gives one equation in three digits. Gather it as

$$99a + (10b - b^2) = c^3 - c,$$

and the two facts from the first paragraph slot straight in: the b-term is trapped between 0 and 25, and both $99a$ and $c^3 - c$ are multiples of 3. The first fact means $c^3 - c$ must sit just above a multiple of 99; the second cuts down the possible values of b. After that there are only a handful of cases to check by hand.

Solution

The two preliminary facts

Completing the square,

$$10k - k^2 = 25 - (5 - k)^2.$$

Since $(5 - k)^2 \geq 0$, we get $10k - k^2 \leq 25$. Since $0 \leq k \leq 9$, we have $-4 \leq 5 - k \leq 5$, so $(5 - k)^2 \leq 25$ and hence $10k - k^2 \geq 0$. (Equality at 0 happens for $k = 0$ and at 25 for $k = 5$.)

Next, $k - 1$, k, $k + 1$ are three consecutive integers, so exactly one of them is a multiple of 3. Hence their product $k(k - 1)(k + 1)$ is divisible by 3. (This holds for every integer k, not just $0 \leq k \leq 9$.)

Finding the numbers

Let $N = 100a + 10b + c$ with digits $1 \leq a \leq 9$ (as $N \geq 100$) and $0 \leq b, c \leq 9$. Then $S = a + b^2 + c^3$, and

$$\begin{aligned} S = N &\iff a + b^2 + c^3 = 100a + 10b + c \\ &\iff 99a + (10b - b^2) = c^3 - c = (c - 1)c(c + 1). \qquad (*) \end{aligned}$$

Ask yourself Why put $10b - b^2$ on the same side as $99a$?

Because that is the combination we have just learned to control. With $t = 10b - b^2$, the first fact says $0 \le t \le 25$, and $99a$ is a multiple of 99 that is much bigger than 25. So $(*)$ says that $c^3 - c$ is a multiple of 99 plus a small "remainder" t, which pins things down very quickly.

Restricting b. By the second fact, 3 divides $c^3 - c = (c-1)c(c+1)$, and 3 divides $99a$. So by $(*)$, 3 divides $t = b(10-b)$. Since $10 - b \equiv 1 - b \pmod 3$, this needs $b \equiv 0$ or $b \equiv 1 \pmod 3$. The possibilities are

b	0	1	3	4	6	7	9
$t = 10b - b^2$	0	9	21	24	24	21	9

so $t \in \{0, 9, 21, 24\}$. (The values $b = 2, 5, 8$ give $t = 16, 25, 16$, none divisible by 3.)

Restricting c. Since $a \ge 1$ and $t \ge 0$, $(*)$ gives $c^3 - c \ge 99$. The values of $c^3 - c$ are

c	0	1	2	3	4	5	6	7	8	9
$c^3 - c$	0	0	6	24	60	120	210	336	504	720

so $c \ge 5$.

The cases. For each $c \ge 5$ we need a multiple $99a$ (with $1 \le a \le 9$) lying in the range $[c^3 - c - 25,\ c^3 - c]$, and then $t = c^3 - c - 99a$ must be one of $0, 9, 21, 24$.

- $c = 5$: $c^3 - c = 120$. The only multiple of 99 in $[95, 120]$ is 99, giving $a = 1$ and $t = 21$, so $b = 3$ or $b = 7$. This gives $N = 135$ and $N = 175$.
- $c = 6$: $c^3 - c = 210$. The only multiple of 99 in $[185, 210]$ is 198, giving $a = 2$ and $t = 12$. But 12 is not one of $0, 9, 21, 24$ (indeed $10b - b^2 = 12$ has no integer root). No solution.
- $c = 7$: $c^3 - c = 336$. The multiples nearby are 297 and 396, neither in $[311, 336]$. No solution.
- $c = 8$: $c^3 - c = 504$. The only multiple of 99 in $[479, 504]$ is 495, giving $a = 5$ and $t = 9$, so $b = 1$ or $b = 9$. This gives $N = 518$ and $N = 598$.
- $c = 9$: $c^3 - c = 720$. The multiples nearby are 693 and 792, neither in $[695, 720]$. No solution.

Every step above is reversible, so these four do satisfy $S = N$; checking directly,

$$135 : 1 + 3^2 + 5^3 = 1 + 9 + 125 = 135, \qquad 175 : 1 + 7^2 + 5^3 = 1 + 49 + 125 = 175,$$
$$518 : 5 + 1^2 + 8^3 = 5 + 1 + 512 = 518, \qquad 598 : 5 + 9^2 + 8^3 = 5 + 81 + 512 = 598.$$

So the numbers with $S = N$ are exactly

$$N = 135, \quad 175, \quad 518, \quad 598.$$

Checks

A brute-force search over all 900 three-digit numbers, computing $a + b^2 + c^3$ for each, finds exactly 135, 175, 518 and 598, agreeing with the case analysis. A neat structural check: in each pair the two values of b add to 10 ($3 + 7$ and $1 + 9$). That is forced, because $t = 10b - b^2$ is unchanged when b is replaced by $10 - b$.

Common slips

- Forgetting that $a \geq 1$, and so admitting "solutions" such as 000 or 001 that are not three-digit numbers.
- Stopping after finding one b for a given a and c: since $t(b) = t(10 - b)$, solutions come in pairs.
- Using only the divisibility by 3 and not the bound $0 \leq t \leq 25$: it is the bound that leaves at most one possible a for each c.

Beyond STEP

Numbers equal to a sum built from powers of their own digits are a classic recreation (the Armstrong or "narcissistic" numbers, such as $153 = 1^3 + 5^3 + 3^3$, are the best-known family). The serious idea underneath is the one used here: bound the size of each side to make the search finite, then use congruences to prune it. The same pairing of size estimates with modular arithmetic drives the solution of many Diophantine equations. Davenport's *The Higher Arithmetic* is an excellent introduction to congruences and their uses.

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